

Software Requirements Specification (SRS)

Project: Hostel Management System

Version: 1.0

Team Members: -

Pradeep - PES2UG24CS357
Pavan kumar H S - PES2UG24CS344
Prabhakar kumar - PES2UG24CS355
Pavithra G C - PES2UG24CS346

Team Number:-11

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1 INTRODUCTION

1.1 Purpose

The purpose of the **Hostel Management System** is to provide a simple and organized system for managing hostel room allocation and student details. The system helps hostel administrators maintain student records, manage hostel rooms, check room availability, and allocate rooms to students efficiently.

The system reduces manual record keeping and makes hostel-related information easier to manage and access.

1.2 Scope

The **Hostel Management System** is designed to manage hostel student details and room allocation. The system allows the administrator to maintain student information, manage hostel and room details, check room availability, and allocate available rooms to students.

The system focuses on providing an organized method for maintaining hostel records and reducing errors associated with manual management.

1.3 Audience

This Software Requirements Specification (SRS) document is intended for:

- Hostel administrators who manage student and room information.
- Students whose hostel and room details are maintained in the system.
- Developers responsible for designing and implementing the system.
- Testers responsible for verifying the system requirements.

- Project team members and faculty who review the mini-project.

1.4 Definitions

Term	Definition
HMS	Hostel Management System
Admin	Authorized person responsible for managing hostel records and room allocation
Student	A student whose hostel and room details are maintained in the system
Hostel	Residential facility provided for students
Room	A hostel accommodation unit assigned to one or more students
Room Allocation	The process of assigning an available hostel room to a student
Room Availability	Information about whether a hostel room has available capacity
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Hostel Management System is a standalone system developed to support the basic management activities of a student hostel. It maintains student details and hostel room information in a structured manner.

The system provides a centralized way for the hostel administrator to manage student records, room details, room availability, and room allocation. It reduces dependency on manual records and helps in maintaining accurate and easily accessible information.

2.2 Major Product Functions

The major functions of the Hostel Management System are:

- Admin login and authentication.
- Adding and maintaining student details.
- Updating student information.
- Managing hostel details.
- Adding and maintaining room details.
- Checking room availability.
- Allocating rooms to students.
- Viewing allocated room details.
- Updating room allocation information.
- Removing or vacating a room allocation.
- Searching for student and room information.
- Maintaining records of available and occupied rooms.

2.3 User Roles and Characteristics

The system mainly involves the following users:

Hostel Administrator

The hostel administrator is the main user of the system. The administrator can manage student details, hostel rooms, room availability, and room allocation. The administrator is expected to have basic knowledge of using a computer and web-based applications.

Student

Students are the beneficiaries of the system. Their details are maintained in the system for hostel management and room allocation purposes. Students may view their relevant hostel and room information depending on the system access provided.

2.4 Operating Environment

The Hostel Management System is intended to operate in the following environment:

- **Operating System:** Windows / Linux
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python
- **Framework:** Django or Flask
- **Database:** MySQL
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox
- **System:** Desktop or laptop computer
- **Network:** Local network or Internet connection, depending on deployment

2.5 Constraints

The following constraints are considered for the system:

- Only authorized users should be allowed to access administrative functions.
- Room allocation should be performed only when a room has availability.
- Student information should be stored accurately in the database.
- The system depends on the availability of the database and application server.
- The system is limited to hostel student and room management activities.
- The system should follow the technologies and development framework selected by the project team.

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 User Interfaces

The Hostel Management System should provide a simple and easy-to-use interface for the users. The main screens of the system include:

- Login page for authorized users.
- Student details page for adding and updating student information.
- Hostel and room management page for maintaining room details.
- Room availability page for viewing available and occupied rooms.
- Room allocation page for assigning rooms to students.
- Student search and details page for finding student records.
- Room details page for viewing allocation and occupancy information.

The interface should provide clear labels, input fields, buttons, and messages so that users can perform the required operations without difficulty.

3.2 Hardware Interfaces

The system does not require any special hardware. It can be accessed using a standard computer or laptop.

The minimum hardware requirements are:

- Computer or laptop.
- Keyboard and mouse.
- Monitor/display.
- Network connection, if the system is deployed over a network.

3.3 Software Interfaces

The Hostel Management System interacts with the following software components:

Software	Purpose
Python	Used for backend development.
Django/Flask	Used to develop the web application.
HTML, CSS, JavaScript	Used for designing the user interface.
MySQL	Used for storing student, hostel, room, and allocation details.
Web Browser	Used to access and interact with the system.
Operating System	Provides the environment required to run the application.

3.4 Communications

The system uses HTTP/HTTPS communication between the web browser and the application server.

The application communicates with the database server to store and retrieve student, hostel, room, and room allocation information.

If the system is deployed on a network, the users should have a reliable network connection to access the application.

4. SYSTEM FEATURES

The Hostel Management System consists of functional requirements that describe the operations performed by the system. Each requirement is assigned a unique ID, priority, source, acceptance criteria, and test case reference.

4.1 Functional Requirements

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-001	The system shall allow the administrator to log in using valid credentials.	Functional	High	Hostel Administrator	TC-001: Login succeeds with valid username and password.	Requires user authentication.
HMS-F-002	The system shall reject login attempts when invalid credentials are entered.	Functional	High	Hostel Administrator	TC-002: Invalid credentials display an appropriate error message.	Depends on authentication module.
HMS-F-003	The system shall allow the administrator to add new student details.	Functional	High	Hostel Administrator	TC-003: A student record is created when valid details are submitted.	Requires student database.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-004	The system shall allow the administrator to view stored student details.	Functional	High	Hostel Administrator	TC-004: The selected student's details are displayed correctly.	Depends on student records.
HMS-F-005	The system shall allow the administrator to update student details.	Functional	High	Hostel Administrator	TC-005: Updated information is saved and displayed correctly.	Student record must exist.
HMS-F-006	The system shall allow the administrator to remove a student record when required.	Functional	Medium	Hostel Administrator	TC-006: The selected student record is removed after confirmation.	Requires existing student record.
HMS-F-007	The system shall allow the administrator to add hostel room	Functional	High	Hostel Administrator	TC-007: Valid room details are successfully stored.	Requires room management module.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	details.					
HMS-F-008	The system shall allow the administrator to view available hostel rooms.	Functional	High	Hostel Administrator	TC-008: Available rooms are displayed with their current availability.	Depends on room records.
HMS-F-009	The system shall allow the administrator to view occupied rooms.	Functional	Medium	Hostel Administrator	TC-009: Occupied rooms are displayed correctly.	Depends on allocation records.
HMS-F-010	The system shall allow the administrator to allocate an available room to a student.	Functional	High	Hostel Administrator	TC-010: A selected available room is assigned to the student successfully.	Student and room records must exist.
HMS-F-011	The system shall prevent allocation of a room when there is no	Functional	High	Hostel Administrator	TC-011: The system prevents allocation when the	Depends on room capacity.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	available capacity.				room is full.	
HMS-F-012	The system shall display the room allocated to a student.	Functional	High	Hostel Administrator / Student	TC-012: The student's current room details are displayed correctly.	Requires successful room allocation.
HMS-F-013	The system shall allow the administrator to update a student's room allocation.	Functional	Medium	Hostel Administrator	TC-013: The student's room is changed and the new allocation is stored.	Existing allocation required.
HMS-F-014	The system shall allow the administrator to vacate a student's allocated room.	Functional	Medium	Hostel Administrator	TC-014: The allocation is removed and the room becomes available.	Requires an existing allocation.
HMS-F-015	The system shall allow the	Functional	Medium	Hostel Administrator	TC-015: The matching	Depends on student database.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	administrator to search for student records using relevant student information.				student record is displayed for a valid search.	
HMS-F-016	The system shall maintain the current availability status of hostel rooms after allocation or vacation.	Functional	High	Hostel Administrator	TC-016: Room availability is updated correctly after allocation or vacation.	Depends on room allocation records.
HMS-F-017	The system shall allow the administrator to view the details of a particular hostel room.	Functional	Medium	Hostel Administrator	TC-017: The selected room's capacity, occupancy, and allocation details are displayed.	Requires room records.

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality, performance, reliability, usability, and security aspects of the Hostel Management System. These requirements specify how the system should perform its functions.

5.1 Non-Functional Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case Ref
HMS-NFR-001	The system should provide a simple and user-friendly interface so that users can perform common operations easily.	Usability	High	TC-NFR-001: User should be able to perform basic operations without difficulty.
HMS-NFR-002	The system should respond to normal user requests within 3 seconds under normal operating conditions.	Performance	High	TC-NFR-002: Response time is measured for common operations and should be within 3 seconds.
HMS-NFR-003	The system should maintain accurate and consistent student, room, and allocation information.	Reliability	High	TC-NFR-003: Stored information should remain consistent after add, update, allocation, and vacation operations.
HMS-NFR-004	The system should be available during normal hostel administrative working hours.	Availability	Medium	TC-NFR-004: The system should be accessible during the defined operating period.
HMS-NFR-	The system should be maintainable so that	Maintainability	Medium	TC-NFR-005: A module can be

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case Ref
005	changes and updates can be made without affecting unrelated functions.			modified or updated without causing failures in other modules.
HMS-NFR-006	The system should support multiple student records without significant degradation in normal operations.	Scalability	Medium	TC-NFR-006: The system should continue to perform normally as the number of student records increases.

5.2 Security

Security requirements are included to protect student information and prevent unauthorized access to hostel records.

5.2.1 Security Objectives

The main security objectives of the Hostel Management System are:

1. **Confidentiality:** Student and hostel information should be accessible only to authorized users.
2. **Integrity:** Student, room, and allocation records should be protected from unauthorized modification or deletion.
3. **Authentication:** Only authorized users should be allowed to access administrative functions.

5.2.2 Security Requirements

Req ID	Security Requirement	Priority	Acceptance Criteria / Test Case Ref
HMS-SEC-001	The system shall require users to provide valid login credentials before accessing administrative functions.	High	TC-SEC-001: Access is granted only with valid credentials.

Req ID	Security Requirement	Priority	Acceptance Criteria / Test Case Ref
HMS-SEC-002	The system shall reject invalid username or password combinations.	High	TC-SEC-002: Invalid credentials should not provide access to the system.
HMS-SEC-003	The system shall restrict student and room management operations to authorized administrators.	High	TC-SEC-003: Unauthorized users cannot access administrative operations.
HMS-SEC-004	The system shall protect stored student and hostel information from unauthorized modification or deletion.	High	TC-SEC-004: Unauthorized modification or deletion attempts are prevented.
HMS-SEC-005	The system shall terminate or restrict access after logout so that protected pages cannot be accessed without authentication.	Medium	TC-SEC-005: After logout, protected pages require authentication again.
HMS-SEC-006	The system shall validate user input before storing student and room information in the database.	Medium	TC-SEC-006: Invalid or incomplete input is rejected before being stored.

6. QUALITY ATTRIBUTES & ACCEPTANCE TESTS

6.1 Quality Attributes

The Hostel Management System should satisfy the following quality attributes to ensure that the system is reliable and easy to use.

Quality Attribute	Requirement
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Quality Attribute	Requirement
Usability	The system should have a simple interface that allows the administrator to perform hostel management activities easily.
Performance	The system should provide a response to normal user operations within the specified response time under normal conditions.
Reliability	The system should maintain accurate student, room, and allocation records during normal operation.
Security	The system should prevent unauthorized users from accessing administrative functions and protected information.
Maintainability	The system should be structured into manageable modules so that changes can be made with minimal impact on other functions.
Scalability	The system should be able to handle an increasing number of student and room records without major performance issues.

6.2 Acceptance Tests

Acceptance tests are used to verify whether the Hostel Management System satisfies the specified functional and non-functional requirements.

Test Case ID	Test Description	Expected Result	Requirement Ref.
AT-001	Enter valid administrator login credentials.	Administrator should be successfully logged in.	HMS-F-001
AT-002	Enter invalid login credentials.	System should reject the login and display an appropriate message.	HMS-F-002

Test Case ID	Test Description	Expected Result	Requirement Ref.
AT-003	Add a new student with valid details.	Student details should be stored successfully.	HMS-F-003
AT-004	Search for an existing student.	Correct student details should be displayed.	HMS-F-004, HMS-F-015
AT-005	Update an existing student's information.	Updated information should be saved correctly.	HMS-F-005
AT-006	Add valid hostel room details.	Room details should be stored successfully.	HMS-F-007
AT-007	View available rooms.	Available rooms should be displayed correctly.	HMS-F-008
AT-008	Allocate an available room to a student.	The room should be assigned successfully.	HMS-F-010
AT-009	Attempt to allocate a full room.	System should prevent the allocation.	HMS-F-011
AT-010	View a student's allocated room.	Correct room information should be displayed.	HMS-F-012
AT-011	Change a student's room allocation.	New room allocation should be stored correctly.	HMS-F-013
AT-012	Vacate an allocated room.	Allocation should be removed and room availability should be updated.	HMS-F-014, HMS-F-016
AT-013	Try to access an administrative function without logging in.	System should prevent unauthorized access.	HMS-SEC-001, HMS-SEC-003

Test Case ID	Test Description	Expected Result	Requirement Ref.
AT-014	Measure response time for normal operations.	System should respond within the specified time under normal conditions.	HMS-NFR-002
AT-015	Modify student information with valid data.	Updated information should remain accurate and consistent.	HMS-NFR-003
AT-016	Logout and try to access a protected page.	System should require authentication again.	HMS-SEC-005

6.3 Exit Criteria

The Hostel Management System will be considered ready for acceptance when:

- All high-priority functional requirements have been implemented and verified.
- The defined acceptance tests are successfully completed.
- No critical security issues remain unresolved.
- The system satisfies the specified non-functional requirements under normal operating conditions.
- Student, room, and room allocation information is stored and retrieved correctly.
- The Requirements Traceability Matrix confirms that the required test cases have been completed successfully.

7. UML USE-CASE DIAGRAM

The UML Use-Case Diagrams represent the interaction between the users and the Hostel Management System. The diagrams show the main activities that can be performed by the users of the system.

7.1 Use-Case Diagram 1 – Administrator

The first use-case diagram represents the activities performed by the **Hostel Administrator**.

Actor:

- Hostel Administrator

Main Use Cases:

- Login
- Manage Student Details
- Add Student
- View Student Details
- Update Student Details
- Remove Student
- Manage Room Details
- Add Room
- View Room Details
- Check Room Availability
- Allocate Room
- Update Room Allocation
- Vacate Room
- Search Student Records
- Logout

Figure Use-Case Diagram – Hostel Administrator



7.2 Use-Case Diagram 2 – Student

The second use-case diagram represents the activities related to the **Student**.

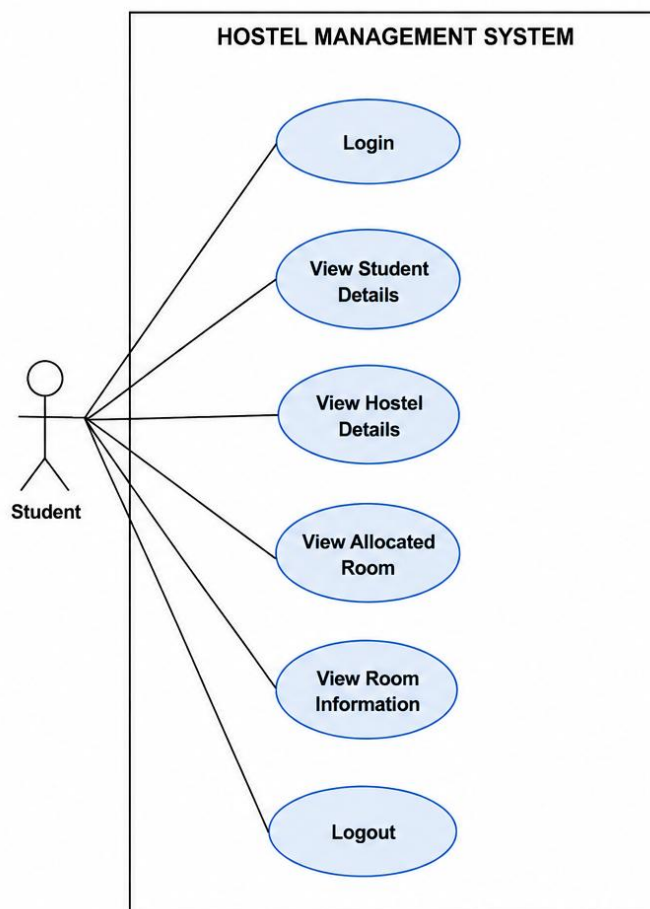
Actor:

- Student

Main Use Cases:

- Login
- View Student Details
- View Hostel Details
- View Allocated Room
- View Room Information
- Logout

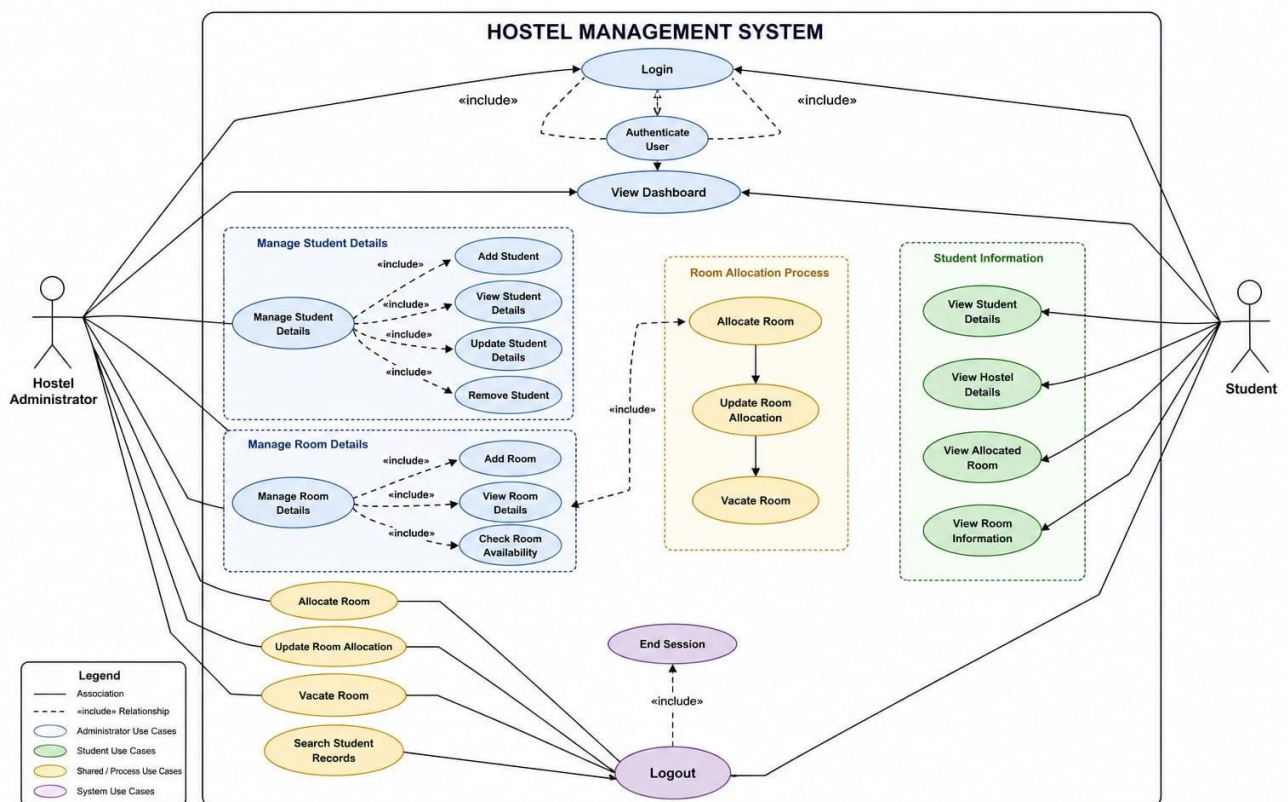
Figure 7.2: Use-Case Diagram – Student



7.3 Actors and Use-Case Summary

Actor	Use Cases
Hostel Administrator	Login, manage student details, manage room details, check room availability, allocate room, update allocation, vacate room, search records, logout
Student	Login, view student details, view hostel details, view allocated room, view room information, logout

The overall Usecase Digram:-



The use-case diagrams provide a high-level view of how the users interact with the Hostel Management System and help in understanding the functional requirements of the system.

8. REQUIREMENTS TRACEABILITY MATRIX (RTM)

The Requirements Traceability Matrix (RTM) is used to establish a connection between the requirements of the Hostel Management System and their corresponding system modules and test cases. It helps to verify that each requirement is considered during development and testing.

Req ID	Requirement Short	Section Ref.	Module	Test Case(s)	Status	Comments
HMS-F-001	Administrator login	4.1	Authentication	TC-001	N	To be tested
HMS-F-002	Invalid login handling	4.1	Authentication	TC-002	N	To be tested
HMS-F-003	Add student details	4.1	Student Management	TC-003	N	To be tested
HMS-F-004	View student details	4.1	Student Management	TC-004	N	To be tested
HMS-F-005	Update student details	4.1	Student Management	TC-005	N	To be tested
HMS-F-006	Remove student record	4.1	Student Management	TC-006	N	To be tested
HMS-F-007	Add room details	4.1	Room Management	TC-007	N	To be tested
HMS-F-008	View available rooms	4.1	Room Management	TC-008	N	To be tested
HMS-F-009	View occupied rooms	4.1	Room Management	TC-009	N	To be tested
HMS-F-010	Allocate room	4.1	Room Allocation	TC-010	N	To be tested

Req ID	Requirement Short	Section Ref.	Module	Test Case(s)	Status	Comments
HMS-F-011	Prevent allocation of full room	4.1	Room Allocation	TC-011	N	To be tested
HMS-F-012	View allocated room	4.1	Room Allocation	TC-012	N	To be tested
HMS-F-013	Update room allocation	4.1	Room Allocation	TC-013	N	To be tested
HMS-F-014	Vacate room	4.1	Room Allocation	TC-014	N	To be tested
HMS-F-015	Search student records	4.1	Student Management	TC-015	N	To be tested
HMS-F-016	Update room availability	4.1	Room Management	TC-016	N	To be tested
HMS-F-017	View room details	4.1	Room Management	TC-017	N	To be tested
HMS-NFR-001	User-friendly interface	5.1	User Interface	TC-NFR-001	N	To be tested
HMS-NFR-002	Response time	5.1	System Performance	TC-NFR-002	N	To be tested
HMS-NFR-003	Data accuracy and consistency	5.1	Database	TC-NFR-003	N	To be tested
HMS-NFR-	System availability	5.1	System	TC-NFR-	N	To be tested

Req ID	Requirement Short	Section Ref.	Module	Test Case(s)	Status	Comments
004				004		
HMS-NFR-005	Maintainability	5.1	System Architecture	TC-NFR-005	N	To be tested
HMS-NFR-006	Scalability	5.1	Database / System	TC-NFR-006	N	To be tested
HMS-SEC-001	User authentication	5.2.2	Authentication	TC-SEC-001	N	To be tested
HMS-SEC-002	Invalid credentials rejection	5.2.2	Authentication	TC-SEC-002	N	To be tested
HMS-SEC-003	Admin access restriction	5.2.2	Authorization	TC-SEC-003	N	To be tested
HMS-SEC-004	Data protection	5.2.2	Database / Security	TC-SEC-004	N	To be tested
HMS-SEC-005	Protected access after logout	5.2.2	Authentication	TC-SEC-005	N	To be tested
HMS-SEC-006	Input validation	5.2.2	Data Validation	TC-SEC-006	N	To be tested

Status Legend

- **N – Not Tested:** Requirement has not yet been tested.
- **P – Passed:** Requirement has been implemented and successfully verified.
- **A – Action Required:** Requirement needs further implementation or correction.

DIVISION OF WORK

The responsibilities for preparing the Software Requirements Specification (SRS) document were divided equally among the four team members. Each member contributed to the preparation, review, and finalization of the document.

Sl. No.	Team Member	Work Assigned
1	Pavithra G C	Introduction, Overall Description, and review of the SRS document, integrated the sections
2	Pavan Kumar H S	External Interface Requirements and Functional Requirements
3	Prabhakar Kumar	Non-Functional Requirements, Security Requirements, and Quality Attributes
4	Pradeep	UML Use-Case Diagrams, Acceptance Tests, and Requirements Traceability Matrix (RTM)

Team Contribution

All four team members participated in reviewing the complete SRS document and making necessary corrections before submission. The responsibilities were distributed to ensure equal contribution towards the deliverable.